

LSSO: Learnable Low-Rank Sylvester Solves for Efficient Bidirectional Encoders

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Abstract

We introduce LSSO, a learnable low-rank global-solve token mixer for bidirectional encoders. Instead of explicit pairwise value routing, LSSO constructs an input-dependent positive-shifted low-rank relation field and solves

$$(\mu I + \gamma U(X)U(X)^\top)Y = C(X).$$

The current operator is a Sylvester-style solve, corresponding to the left-sided special case of the more general equation $AY + YB = C$. We show that the operator is well posed under positive parameterization, admits an efficient Woodbury implementation, has a bounded fixed- U spectral response, and is equivalent to minimizing a strongly convex low-rank correction energy. Experiments on retrieval, MS MARCO to BEIR transfer, vision encoders, ablations, rank pruning, and sequence scaling show that LSSO can provide competitive bidirectional encoder performance with substantially lower mixer-only MACs. Boundary studies further clarify that LSSO is not an attention approximation and is not yet a universal replacement for attention in all generative settings.

1 Introduction

Self-attention has become the default token mixer for bidirectional encoders. Given query, key, and value projections, it produces

$$Y_{\text{attn}} = \text{softmax}(QK^\top / \sqrt{d})V,$$

which can be interpreted as input-dependent pairwise routing. This routing view is expressive, but the score and value-routing terms scale quadratically with sequence length. Many efficient attention methods reduce this cost by approximating or sparsifying the attention matrix. LSSO takes a different route: it replaces pairwise value routing with an input-dependent global linear solve.

For an input sequence $X \in \mathbb{R}^{B \times N \times D}$, LSSO learns a low-rank relation feature $U(X)$ and a content state $C(X)$. For each head it solves

$$(\mu I_N + \gamma U U^\top)Y = C, \quad \mu > 0, \quad \gamma \geq 0,$$

and projects the result back to the model dimension. The matrix $U U^\top$ is positive semidefinite, while individual similarities $u_i^\top u_j$ may be positive or negative. Thus the system

defines a stable global relation field without requiring an explicit dense routing matrix.

The guiding interpretation is:

$$\begin{aligned} \text{LSSO} &= \text{isotropic content pass-through} \\ &\quad - \text{low-rank global correction.} \end{aligned}$$

This differs from attention: attention learns a row-stochastic routing matrix, whereas LSSO learns a positive-definite inverse filter.

Our contributions are:

1. We propose LSSO, a learnable Sylvester-style low-rank global solve operator for bidirectional encoder token mixing.
2. We derive a Woodbury implementation that reduces the solve to an $r \times r$ system, where r is the rank.
3. We prove fixed- U well-posedness, spectral boundedness, conditioning control, and an energy minimization equivalence.
4. We validate LSSO on retrieval, transfer, vision encoders, ablations, rank pruning, and long-sequence scaling.
5. We state the limitations explicitly: LSSO is a bidirectional encoder mixer, not an attention approximation, a causal decoder drop-in, or a universal diffusion backbone.

2 Related Work

Efficient attention. Linformer [9], Performer [3], Nyströmformer [10], Longformer [2], and BigBird [11] reduce the cost of self-attention through low-rank projections, kernel approximations, landmark approximations, or sparse attention patterns. These methods are best understood as efficient attention or attention-approximation mechanisms. LSSO does not attempt to approximate the softmax attention matrix; it defines a different input-dependent inverse-filter token mixer.

Token mixers beyond attention. MLP-Mixer [8], gMLP [6], FNet [5], Hyena [7], and state-space models such as Mamba [4] show that token mixing need not be implemented as dense attention. LSSO adds another point in this design space: an implicit low-rank global solve specialized for non-causal encoders.

Implicit layers and learned solves. Deep equilibrium models [1], implicit layers, graph smoothing, and regularized least-squares methods use linear or nonlinear solves as model components. LSSO is much simpler than a general equilibrium layer: it performs one closed-form, positive-shifted low-rank correction whose matrix is generated from the input.

3 Method

3.1 Encoder Block

We use LSSO in the token-mixer position of a pre-normalized encoder block:

$$X' = X + \text{LSSO}(\text{LN}(X)), \quad X_{\text{out}} = X' + \text{MLP}(\text{LN}(X')).$$

All other components are kept unchanged in controlled comparisons: layer normalization, residual paths, feed-forward layers, positional embeddings, and task heads.

3.2 LSSO Definition

For each head, let

$$U = XW_u \in \mathbb{R}^{B \times N \times r}, \quad C = XW_c \in \mathbb{R}^{B \times N \times d_h}.$$

We optionally RMS-normalize U along its rank dimension and solve

$$(\mu I_N + \gamma U U^\top) Y = C, \quad (1)$$

where

$$\mu = \text{softplus}(\theta_\mu) + \epsilon, \quad \gamma = \gamma_{\max} \sigma(\theta_\gamma).$$

The default experiments use $\gamma_{\max} = 0.3$ and $\theta_\gamma = -4$, which gives an initial γ/μ of roughly 7.8×10^{-3} when $\theta_\mu = 0$. Smaller settings such as $\gamma_{\max} = 0.1, \theta_\gamma = -6$ make the global correction nearly vanish at initialization and were empirically too conservative for the main runs.

The head outputs are concatenated and projected:

$$\text{LSSO}(X) = YW_o.$$

3.3 Woodbury Implementation

Directly inverting the $N \times N$ matrix in Eq. 1 is unnecessary. By the Woodbury identity,

$$Y = \mu^{-1} C - \mu^{-1} U (I_r + \gamma \mu^{-1} U^\top U)^{-1} (\gamma \mu^{-1} U^\top C). \quad (2)$$

This form is continuous at $\gamma = 0$ and only solves an $r \times r$ system. The core solve terms cost

$$O(Nr^2 + Nr d_h + r^3)$$

per head. Including projections, the mixer-only MAC formula used in our tables is

$$ND(Hr + D) + ND^2 + HNr^2 + 2NrD + Hr^3.$$

For MHA we count QKV projections, attention scores, attention-value product, and output projection:

$$4ND^2 + 2N^2D.$$

4 Theoretical Properties

The following results analyze the operator for fixed U . They do not claim global convergence or stability of an entire trained deep network; those are empirical questions beyond the scope of this operator analysis.

Proposition 1 (Well-posedness). *Let $A = \mu I_N + \gamma U U^\top$ with $\mu > 0$ and $\gamma \geq 0$. Then A is symmetric positive definite. Therefore, for any C , the system $AY = C$ has a unique solution.*

Proof. For any nonzero z ,

$$z^\top A z = \mu \|z\|_2^2 + \gamma \|U^\top z\|_2^2 \geq \mu \|z\|_2^2 > 0. \quad \square$$

Proposition 2 (Spectral shrinkage). *Let $U U^\top = Q \Lambda Q^\top$ be an eigendecomposition over the nonzero eigen-directions of $U U^\top$, with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_k)$ and $\lambda_i > 0$. Let $P_\perp = I - Q Q^\top$. Then*

$$A^{-1} = Q \text{diag}\left(\frac{1}{\mu + \gamma \lambda_i}\right) Q^\top + \mu^{-1} P_\perp.$$

Thus LSSO applies a learnable inverse spectral filter along the span of U and isotropic scaling on the orthogonal complement. In this sense it is a data-dependent inverse filter rather than a row-stochastic routing matrix.

Proposition 3 (Bounded fixed- U operator norm). *For fixed U ,*

$$\|\text{LSSO}(C, U)\|_F \leq \mu^{-1} \|C\|_F.$$

With an output projection,

$$\|W_o \text{LSSO}(C, U)\|_F \leq \|W_o\|_2 \mu^{-1} \|C\|_F.$$

Proof. Since $A \succeq \mu I$, $\lambda_{\min}(A) \geq \mu$ and $\|A^{-1}\|_2 \leq \mu^{-1}$. The Frobenius bound follows from $Y = A^{-1}C$. $\square$

Proposition 4 (Conditioning control). *For $A = \mu I + \gamma U U^\top$,*

$$\kappa_2(A) \leq 1 + \frac{\gamma}{\mu} \|U\|_2^2.$$

This explains the diagnostics used in our experiments. The ratio γ/μ directly scales the possible condition-number increase; effective rank measures how much of the rank subspace is used; and correction ratio measures how strongly the low-rank correction participates in the output. RMS normalization of U helps control the scale of U , although it is not by itself a dimension-free spectral-norm guarantee.

Theorem 1 (Energy minimization equivalence). *For $\mu > 0$ and $\gamma \geq 0$, the LSSO output is the unique minimizer of*

$$\mathcal{E}(Y) = \frac{\mu}{2} \|Y\|_F^2 + \frac{\gamma}{2} \|U^\top Y\|_F^2 - \langle C, Y \rangle.$$

Proof. The Hessian of $\mathcal{E}$ is $\mu I + \gamma U U^\top$, which is positive definite by the well-posedness proposition. Setting the gradient to zero gives $(\mu I + \gamma U U^\top)Y = C$. $\square$

Equivalently, up to an additive constant,

$$\mathcal{E}(Y) = \frac{\mu}{2} \|Y - \mu^{-1}C\|_F^2 + \frac{\gamma}{2} \|U^\top Y\|_F^2.$$

The content projection proposes the local state $\mu^{-1}C$; the learned low-rank term applies a global correction.

Proposition 5 (No-global degeneration). *If $\gamma = 0$, then*

$$\text{LSSO}(C, U) = \mu^{-1}C.$$

No token mixing occurs except for pointwise projections.

5 Experiments

5.1 Setup

All main retrieval and vision comparisons use randomly initialized encoders. Tokenizers may be reused, but encoder weights are trained from scratch. LSSO uses $\gamma_{\max} = 0.3$ and $\theta_\gamma = -4$ unless stated otherwise. Tables report mixer-only MACs to isolate the cost of replacing the token mixer; FFN cost is intentionally not folded into the main MAC column.

5.2 Retrieval Main Results

Table 1 reports 3-seed retrieval results on FIQA, NFCorpus, and SciFact. The baseline is a BERT-style encoder with MHA. Nyströmformer uses the official attention module. LSSO-r16 and LSSO-r32 keep the same model dimension, depth, heads, optimizer, and training schedule.

LSSO-r16 reduces mixer MACs by 66.8% relative to MHA, while LSSO-r32 reduces them by 57.7%. Both retain or improve MHA-level retrieval quality across the three datasets. Nyströmformer is a strong retrieval baseline in this setup, but its observed throughput is lower than both MHA and LSSO in our runs.

5.3 MS MARCO to BEIR Transfer

To test whether the result is limited to small supervised retrieval datasets, we pretrain on MS MARCO and evaluate zero-shot on BEIR-style tasks. Macro averages over FIQA, NFCorpus, SciFact, ArguAna, and TREC-COVID are shown in Table 2.

The LSSO models preserve MHA-level transfer under substantially lower mixer cost. In this setting the official Nyströmformer baseline is cheaper than MHA but trails the MHA and LSSO macro averages.

5.4 Ablations and Diagnostics

Table 3 aligns the empirical ablations with the theory. The no-global variant sets $\gamma = 0$, reducing the operator to $\mu^{-1}C$ and removing token mixing. Removing RMS normalization weakens the global correction, especially on SciFact. Smaller ranks expose the capacity of the learned relation subspace.

Figure 1 plots trained layer-wise diagnostics. All tasks show nonzero correction ratios, confirming that the trained operator does not collapse to the local projection. Effective rank is task dependent: SciFact uses much more of the available rank-32 subspace than FIQA, NFCorpus, or CIFAR-100.

5.5 Sequence Scaling

We benchmark single-layer token mixers at sequence lengths from 128 to 2048 with batch size 8, $D = 256$, 8 heads, bf16 autocast, and no optimizer update. Figure 2 shows forward latency, forward+backward latency, incremental peak memory, and theoretical mixer MACs.

At $N = 2048$, LSSO-r16 uses 0.357G mixer MACs compared with 2.684G for MHA, an 86.7% reduction, and reduces forward+backward latency from 2.61 ms to 1.88 ms. LSSO-r32 uses 0.453G MACs and takes 1.95 ms. Peak memory is closer to optimized MHA than the dense $O(N^2)$ formula would suggest because current PyTorch MHA uses memory-efficient Flash/SDPA kernels. Thus this benchmark supports the arithmetic scaling and long-sequence latency claim, not a universal memory-savings claim.

5.6 Vision and Diffusion Boundary Results

Table 4 reports vision encoder results. On CIFAR-100, LSSO gives a cost-accuracy trade-off but does not dominate the official Nyströmformer baseline. On ImageNet-100, MHA remains stronger at one seed, while LSSO reduces mixer MACs substantially. These results support LSSO as a general encoder mixer candidate rather than a universal replacement.

We also ran a latent diffusion boundary experiment on ImageNet-100 latent tokens. It reports noise-prediction validation MSE only, without FID/KID or sample-quality claims. Pure LSSO retains much of the MHA optimization quality with lower theoretical mixer MACs, but it is not yet a direct replacement for attention in diffusion generation.

Table 1: Retrieval main results. Models are randomly initialized BERT-style encoders with $D = 256$, depth 8, heads 8, and document length 512. MACs are mixer-only document-side MACs.

Dataset	Model	Params (M)	Mixer MACs (G)	R@10	MRR@10
FIQA	MHA	14.33	2.147	0.2145 ± 0.0135	0.0987 ± 0.0032
FIQA	Nyströmformer	14.33	1.301	0.2428 ± 0.0039	0.1190 ± 0.0071
FIQA	LSSO-r16	13.54	0.713	0.2258 ± 0.0125	0.1126 ± 0.0061
FIQA	LSSO-r32	13.80	0.908	0.2387 ± 0.0183	0.1150 ± 0.0035
NFCorpus	MHA	14.33	2.147	0.5439 ± 0.0187	0.3809 ± 0.0145
NFCorpus	Nyströmformer	14.33	1.301	0.5841 ± 0.0125	0.4323 ± 0.0132
NFCorpus	LSSO-r16	13.54	0.713	0.5686 ± 0.0125	0.4121 ± 0.0036
NFCorpus	LSSO-r32	13.80	0.908	0.5841 ± 0.0036	0.4333 ± 0.0220
SciFact	MHA	14.33	2.147	0.6789 ± 0.0190	0.5994 ± 0.0138
SciFact	Nyströmformer	14.33	1.301	0.7267 ± 0.0133	0.6313 ± 0.0033
SciFact	LSSO-r16	13.54	0.713	0.7022 ± 0.0107	0.6214 ± 0.0119
SciFact	LSSO-r32	13.80	0.908	0.7089 ± 0.0117	0.6247 ± 0.0080

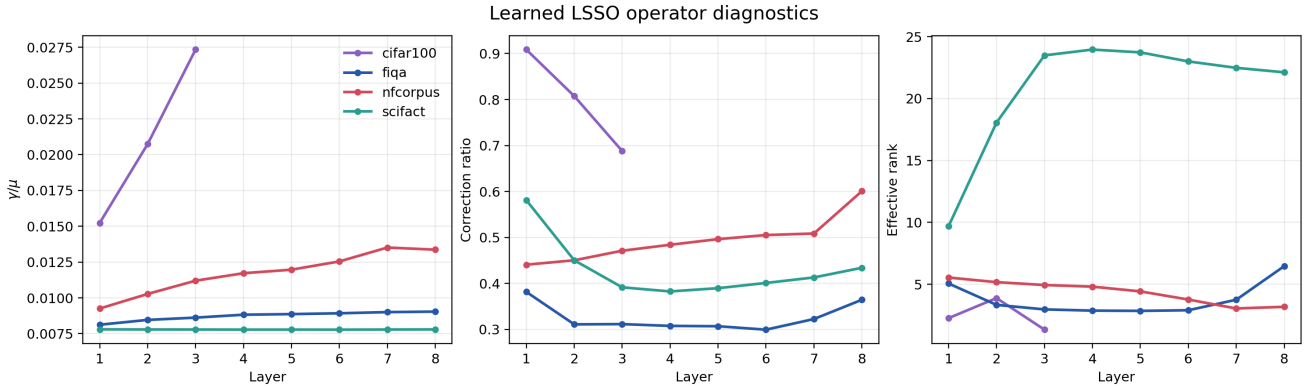


Figure 1: Layer-wise LSSO diagnostics from trained LSSO-r32 checkpoints.

Table 2: MS MARCO to BEIR transfer macro average over five datasets.

Model	MACs (G)	nDCG@10	Recall@100
MHA	2.147	0.22945	0.36061
Nyströmformer	1.270	0.20523	0.33456
LSSO-r16	0.713	0.22973	0.35889
LSSO-r32	0.908	0.22940	0.36367

Table 3: Retrieval ablations. Values are mean $\pm$ standard deviation over three seeds.

Dataset	Variant	R@10	MRR@10	γ/μ	Correction ratio
FIQA	no-global r32	0.2052 ± 0.0137	0.0911 ± 0.0019	0.0000	0.0000
FIQA	fixed μ/γ r32	0.2428 ± 0.0126	0.1136 ± 0.0028	0.0078	0.2130
FIQA	no U RMS norm r32	0.2160 ± 0.0101	0.1007 ± 0.0008	0.0085	0.2050
FIQA	full r8	0.2088 ± 0.0208	0.0923 ± 0.0139	0.0100	0.1059
FIQA	full r4	0.2042 ± 0.0045	0.0979 ± 0.0036	0.0109	0.0742
SciFact	no-global r32	0.6767 ± 0.0120	0.5811 ± 0.0058	0.0000	0.0000
SciFact	fixed μ/γ r32	0.7089 ± 0.0102	0.6088 ± 0.0066	0.0078	0.2631
SciFact	no U RMS norm r32	0.6933 ± 0.0133	0.5972 ± 0.0150	0.0078	0.0471
SciFact	full r8	0.7067 ± 0.0173	0.6118 ± 0.0064	0.0078	0.1149
SciFact	full r4	0.7022 ± 0.0069	0.6146 ± 0.0107	0.0078	0.0766

Table 4: Vision encoder results. CIFAR-100 is 3 seeds; ImageNet-100 is one seed.

Dataset	Model	MACs (G)	Top-1	Top-5
CIFAR-100	MHA	0.0665	55.40 ± 0.27	–
CIFAR-100	Nyströmformer	0.0389	59.98 ± 0.58	–
CIFAR-100	LSSO-r16	0.0249	54.79 ± 0.14	–
CIFAR-100	LSSO-r32	0.0385	55.38 ± 0.18	–
ImageNet-100	MHA	0.520	82.26	95.54
ImageNet-100	LSSO-r16	0.137	80.54	94.88
ImageNet-100	LSSO-r32	0.174	80.58	94.76

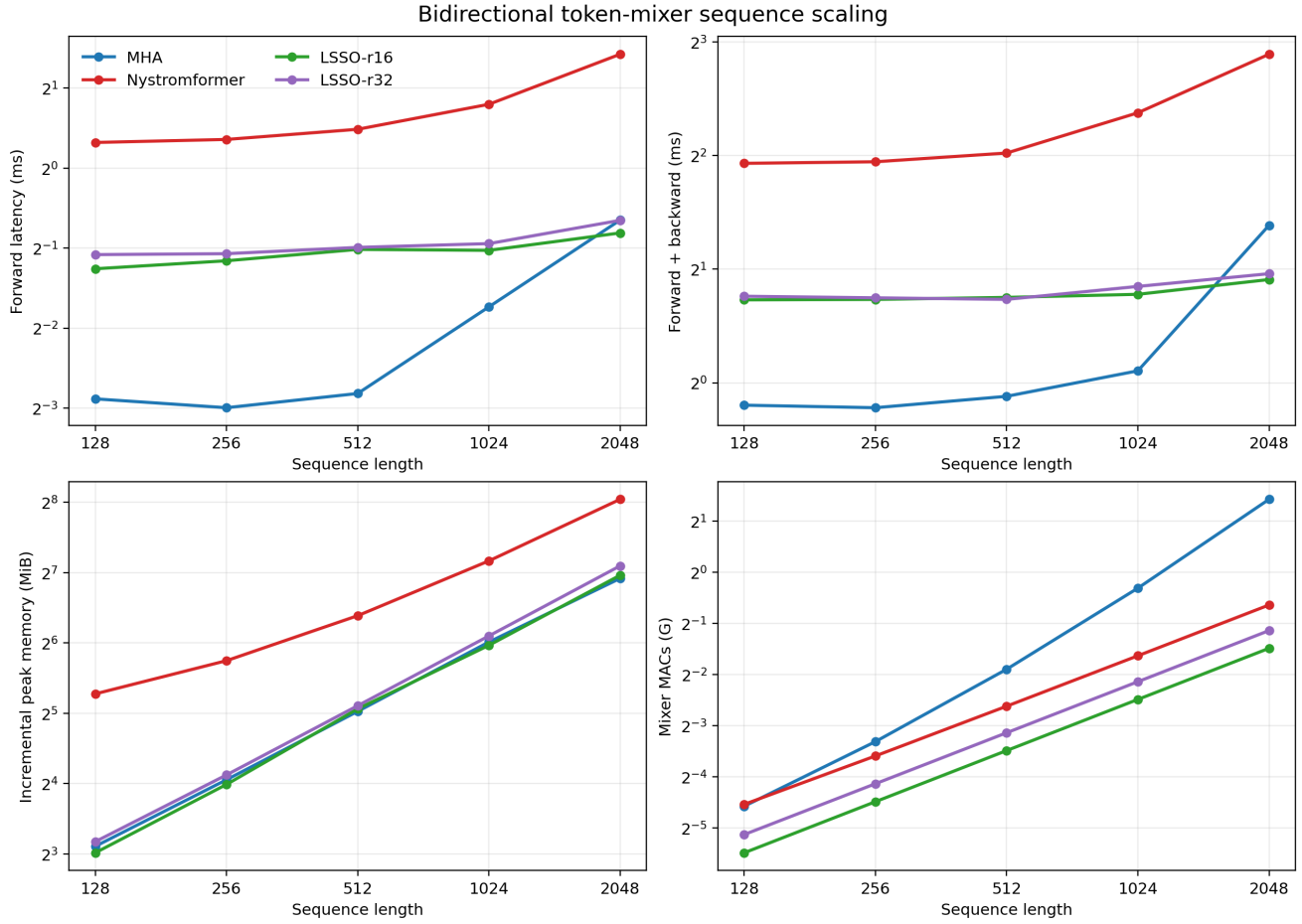


Figure 2: Sequence scaling benchmark on an NVIDIA RTX 5070 Ti. MACs are theoretical per-sample single-layer mixer MACs; latency and memory use batch 8.

6 Discussion

What LSSO is. LSSO is a low-rank global solve, a stable implicit correction operator for fixed U , and a bidirectional encoder token mixer.

What LSSO is not. LSSO is not an attention approximation. It is not currently a causal decoder-only replacement, because causal masking destroys the simple low-rank Woodbury structure. It is also not yet a universal diffusion backbone.

Why retrieval is a natural fit. Retrieval encoders require global bidirectional semantic compression rather than precise causal next-token routing. A low-rank global correction can capture document-level structure while avoiding explicit all-pairs value routing.

7 Limitations

First, the theory analyzes fixed- U operator properties. It does not prove stability of full end-to-end training dynamics. Second, LSSO is designed for bidirectional encoders; causal and autoregressive extensions require new prefix-restricted solves or cached global states. Third, rank is a real capacity knob: some tasks use only a compact subspace, while others, such as SciFact in our diagnostics, use much more of rank 32. Fourth, vision and diffusion experiments show useful cost-accuracy trade-offs but not universal dominance over attention or strong attention alternatives.

8 Conclusion

We introduced LSSO, a learnable low-rank Sylvester-style solve operator for efficient bidirectional token mixing. The operator is well posed, efficiently computable through Woodbury, spectrally bounded for fixed relation features, conditioned by γ/μ and the scale of U , and equivalent to solving a strongly convex global correction energy. Experiments show strong retrieval and transfer performance under lower mixer-only cost, while boundary studies clarify where LSSO is and is not currently effective.

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A Full Woodbury Derivation

The Woodbury identity gives

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}.$$

Set $A = \mu I_N$, $C = \gamma I_r$, and $V = U^\top$:

$$\begin{aligned} (\mu I + \gamma U U^\top)^{-1} &= \mu^{-1} I \\ &\quad - \mu^{-1} U (\gamma^{-1} I + \mu^{-1} U^\top U)^{-1} U^\top \mu^{-1}. \end{aligned}$$

For implementation it is preferable to avoid γ^{-1} , so we rewrite the solution as

$$Y = \mu^{-1}C - \mu^{-1}U(I + \gamma\mu^{-1}U^{\top}U)^{-1}(\gamma\mu^{-1}U^{\top}C).$$

This expression remains valid at $\gamma = 0$.

B Implementation Notes

The PyTorch implementation computes $U^{\top}U$ and $U^{\top}C$ in batched form, solves the small $r \times r$ system in float32 for numerical stability under fp16/bf16 autocast, and casts the result back to the activation dtype. Diagnostics record γ/μ , effective rank of $U^{\top}U$, and the ratio between the Woodbury correction and the local $\mu^{-1}C$ term.